Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18CSI301]

THIRD SEMESTER B.Tech EXAMINATION, DECEMBER–2021

DATA STRUCTURES USING C

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1.	**a.**	Describe sparse matrix with a suitable example.	**07**
	b.	Explain dynamic memory allocation functions.	**07**
2.	**a.**	Evaluate the postfix expression "2 3 1 * + 9 –" using stack.	**08**
	b.	Write a C function to insert and delete an element in circular queue.	**06**
3.	**a.**	Explain representation of linked list in memory.	**07**
	b.	Write an algorithm to insert a node in any position in doubly linked list.	**07**
		OR	
4.	**a.**	Write an algorithm to perform PUSH and POP operations in a stack.	**08**
	b.	Mention the advantages and disadvantages of circular list.	**06**
5.	**a.**	Discuss array representation of binary tree.	**07**
	b.	Explain Threaded binary tree with an example.	**07**
		OR	
6.	**a.**	Outline different traversal techniques of a binary tree.	**06**
	b.	Construct a C function to insert an element into binary search tree.	**08**
7.	**a.**	List the applications of trees and graphs.	**07**
	b.	Design an algorithm for breadth first search.	**07**
		OR	
8.	**a.**	Discuss Kruskal's algorithm with suitable example.	**08**
	b.	Explain Prim's algorithm with an example.	**06**